

BANKING TRANSACTION DASHBOARD

ABSTRACT

The Banking Transaction Dashboard project is developed using Power BI to analyze banking transactions, customer loans, merchant categories, KYC verification status, and transaction channels. The dashboard converts raw banking data into meaningful insights using interactive charts, graphs, and KPI cards.

This project helps financial institutions monitor transaction activities, analyze customer behavior, improve decision-making, and identify business trends effectively. The dashboard also provides state-wise transaction analysis and customer loan distribution details.

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1. INTRODUCTION

Banks generate large amounts of transaction data every day. Managing and analyzing this data manually is difficult and time-consuming. The Banking Transaction Dashboard project helps convert raw banking transaction data into meaningful visual insights using Power BI.

The dashboard helps users understand transaction performance, customer loan activities, merchant categories, KYC verification status, and transaction channels. Interactive charts and filters improve business analysis and support better decision-making.

This project demonstrates the practical application of data analytics and visualization concepts in the banking sector.

2. OBJECTIVES OF THE PROJECT

The main objective of this project is to analyze banking transactions effectively using Power BI dashboards.

Objectives:

- To analyze banking transaction data
 - To monitor customer loan activities
 - To identify transaction patterns across different states
 - To track customer KYC verification status
 - To analyze transaction channels such as ATM, Mobile App, and Web
 - To provide interactive business insights through visualizations
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3. DATASET DESCRIPTION

The dataset contains banking transaction details such as transaction amount, account balance, EMI amount, loan type, merchant category, transaction channel, and customer KYC status.

Important Columns in the Dataset:

Column Name	Description
transaction_amount	Amount transferred during transaction
account_balance	Customer account balance
emi_amount	Monthly EMI amount
loan_type	Type of customer loan
merchant_category	Business category involved
transaction_channel	Transaction method used
kyc_status	Customer verification status
state	State where transaction occurred

The dataset is cleaned and transformed before creating the dashboard. Proper preprocessing improves data quality and dashboard performance.

4. TOOLS AND TECHNOLOGIES USED

Power BI

Power BI is used for dashboard creation and interactive data visualization.

Power Query

Power Query is used for data cleaning and transformation.

DAX (Data Analysis Expressions)

DAX formulas are used for KPI calculations and business measures.

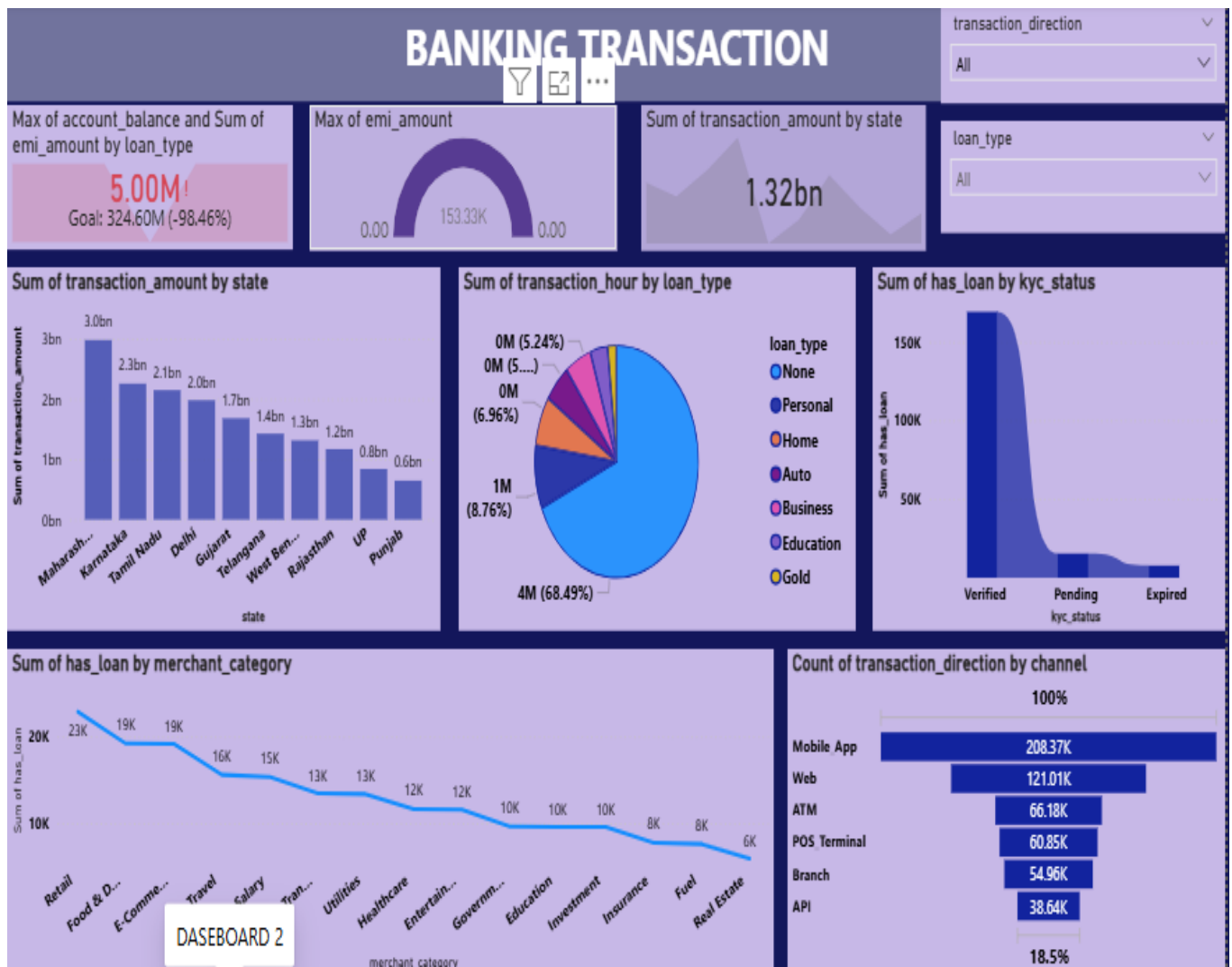
Excel / CSV

Used as the data source for importing transaction data.

Python & Pandas

Used for preprocessing and handling banking transaction data.

5. DASHBOARD EXPLANATION



KPI Cards

The KPI cards displayed at the top of the dashboard provide a quick overview of important banking metrics such as account balance, EMI amount, and total transaction amount. These cards help users easily understand the current financial performance of the bank.

The account balance KPI shows the maximum balance available in customer accounts. The EMI amount card displays the total or average EMI collected from customers. The transaction amount KPI highlights the overall banking transactions performed. KPI cards improve decision-making by presenting key business information in a simple format. They also help management monitor business performance instantly.

State-wise Transaction Analysis

The state-wise transaction analysis bar chart shows the total transaction amount generated from different states. This visualization helps identify which states contribute more to banking transactions. Maharashtra has the highest transaction amount, showing strong banking activity in that region. Punjab has comparatively lower transaction activity.

The chart helps banks understand regional financial performance and customer transaction behavior. It also supports better resource allocation and marketing strategies. State-wise analysis is important for identifying business growth opportunities in different locations.

Loan Type Distribution

The loan type distribution pie chart represents different categories of loans provided by the bank. The chart includes loan types such as Personal Loan, Home Loan, Auto Loan, Business Loan, Education Loan, and Gold Loan.

This visualization helps identify which loan category is most popular among customers. Personal loans appear to have a higher share compared to other loan types. The chart helps banks understand customer borrowing preferences. It also supports loan policy planning and financial product management. Loan analysis improves customer service and business profitability.

KYC Status Analysis

The KYC status analysis chart displays customer verification details such as Verified, Pending, and Expired statuses. This helps banks monitor customer compliance and maintain security standards. Most customers in the dashboard are marked as Verified, showing successful completion of KYC procedures.

Pending and Expired statuses indicate customers who require document updates or verification. This analysis helps reduce fraud risks and ensures banking regulations are followed properly. KYC tracking is essential for customer authentication and financial security. The visualization supports efficient customer record management.

Merchant Category Analysis

The merchant category analysis line chart shows transaction and loan activities across different business sectors. Categories include Retail, Food & Dining, Healthcare, Utilities, Insurance, Real Estate, and Entertainment. Retail and Food & Dining categories have higher transaction volumes compared to others.

This analysis helps banks understand which industries generate more financial activity. It also helps identify customer spending patterns and market demand. Merchant category analysis supports business strategy planning and financial forecasting. The chart provides valuable insights into sector-wise banking performance.

Transaction Channel Analysis

The transaction channel analysis funnel chart displays the number of transactions performed through different banking channels. Channels include Mobile App, Web, ATM, POS Terminal, Branch, and API. The Mobile App channel has the highest number of transactions, indicating strong customer preference for digital banking services.

ATM and Web channels also contribute significantly to banking transactions. Branch and API transactions are comparatively lower. This analysis helps banks improve digital services and customer convenience. It also supports better channel performance monitoring and operational planning.

Filters and Slicers

The dashboard includes interactive filters and slicers that allow users to customize data analysis. Users can filter data based on loan type, transaction direction, and other categories. These filters improve user experience by allowing detailed and focused analysis.

Slicers help management quickly identify trends and compare different business metrics. Interactive filtering makes the dashboard more flexible and user-friendly. It also improves data exploration and report accuracy. Filters are an important feature for dynamic business intelligence dashboards.

6. DATA CLEANING AND TRANSFORMATION

Before building the dashboard, the dataset was cleaned using Power Query and Python. Missing values were handled properly, duplicate records were removed, and incorrect data types were corrected.

Data Cleaning Steps:

- Removed duplicate values
- Handled missing data
- Corrected data types
- Renamed columns

- Created calculated fields

Clean data improves dashboard accuracy and ensures reliable analysis.

7. DAX MEASURES USED

Total Transaction Amount

Total Transaction = SUM(transaction_amount)

Average EMI

Average EMI = AVERAGE(emi_amount)

Loan Count

Loan Count = COUNT(has_loan)

Month-to-Date Calculation

MTD = TOTALMTD([Transaction Amount], Date)

These DAX formulas help calculate KPIs and improve business analysis.

8. ADVANTAGES OF THE DASHBOARD

- Provides real-time banking insights
 - Improves business decision-making
 - Helps identify customer transaction patterns
 - Supports loan and transaction monitoring
 - Improves customer analysis
 - Interactive visualizations simplify complex data
 - Enhances banking operational efficiency
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9. CONCLUSION

The Banking Transaction Dashboard successfully transforms raw banking data into meaningful business insights. Using Power BI visualizations, the dashboard improves transaction analysis, customer verification tracking, and loan monitoring.

The project demonstrates the practical application of Power BI, DAX, and data analytics concepts in the banking sector. It improves operational efficiency and supports better business management.